

BIG DATA SYSTEMS (CC ZG522)

SITUATED LEARNING – PROJECT PROPOSAL

Hadoop-Based Sales Analytics for Business Performance Analysis

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Course: Big Data Systems (CC ZG522)

Date: 11th August 2026

1. PROBLEM STATEMENT AND BUSINESS CONTEXT

As a data analyst, one of the recurring business requirements is to analyze sales data and identify trends that can support business decision-making. Sales transactions contain valuable information about customers, products, regions, categories, sales channels, payment methods, quantities, prices, and revenue.

As the volume of transaction data increases, manually processing such data using conventional tools becomes time-consuming and difficult to scale. A Big Data platform can provide distributed storage and processing capabilities that allow large datasets to be analyzed efficiently.

This project proposes a **Hadoop-based Sales Analytics solution** that stores sales transaction data in HDFS and processes it using MapReduce under YARN.

For the proof of concept, a sample dataset containing **10,000 sales transactions** was prepared. Each transaction contains information such as:

- Transaction ID
- Date
- Customer ID
- Product
- Category
- Region
- Quantity
- Unit Price
- Sales
- Payment Method
- Channel

The main problem addressed by this project is:

How can a scalable Big Data platform be used to analyze sales transactions and identify monthly sales trends and performance across regions, product categories, and sales channels?

The proposed solution automates the aggregation of sales data and produces analytical results that can assist data analysts and business stakeholders in monitoring business performance.

Why This Is a Big Data Problem

The problem can be viewed using the three major characteristics of Big Data:

Volume:

The proof-of-concept dataset contains 10,000 transactions. In a real business environment, sales systems can generate millions of transactions over a longer period. Hadoop provides distributed storage and processing capabilities that can scale to much larger datasets.

Velocity:

Sales transactions may be generated continuously through different channels such as Online, Store, and Distributor. A scalable analytics system should be capable of processing new data regularly.

Variety:

The dataset contains several types of information including dates, customer identifiers, product details, categories, regions, numerical sales values, payment methods, and sales channels. In a production environment, additional sources could include databases, application systems, and other operational data.

Project Objectives

The main objectives of this project are:

1. Store sales transaction data in HDFS.
2. Process sales records using Hadoop MapReduce running on YARN.
3. Calculate monthly sales totals to identify sales trends.
4. Calculate regional sales totals to compare geographical performance.
5. Calculate category-wise sales totals.
6. Calculate channel-wise sales totals.
7. Demonstrate a working Big Data proof of concept using Hadoop tools.
8. Generate preliminary business insights from the processed data.

2. DATA LANDSCAPE AND SOLUTION DESIGN

2.1 Data Source

The proof-of-concept dataset is a CSV file named **sales.csv** containing 10,000 sales transactions.

The dataset contains the following fields:

Field	Description
TransactionID	Unique transaction identifier
Date	Date of the transaction
CustomerID	Customer identifier
Product	Product purchased

Field	Description
Category	Product category
Region	Sales region
Quantity	Quantity purchased
UnitPrice	Price per unit
Sales	Total transaction sales
PaymentMethod	Payment method
Channel	Sales channel

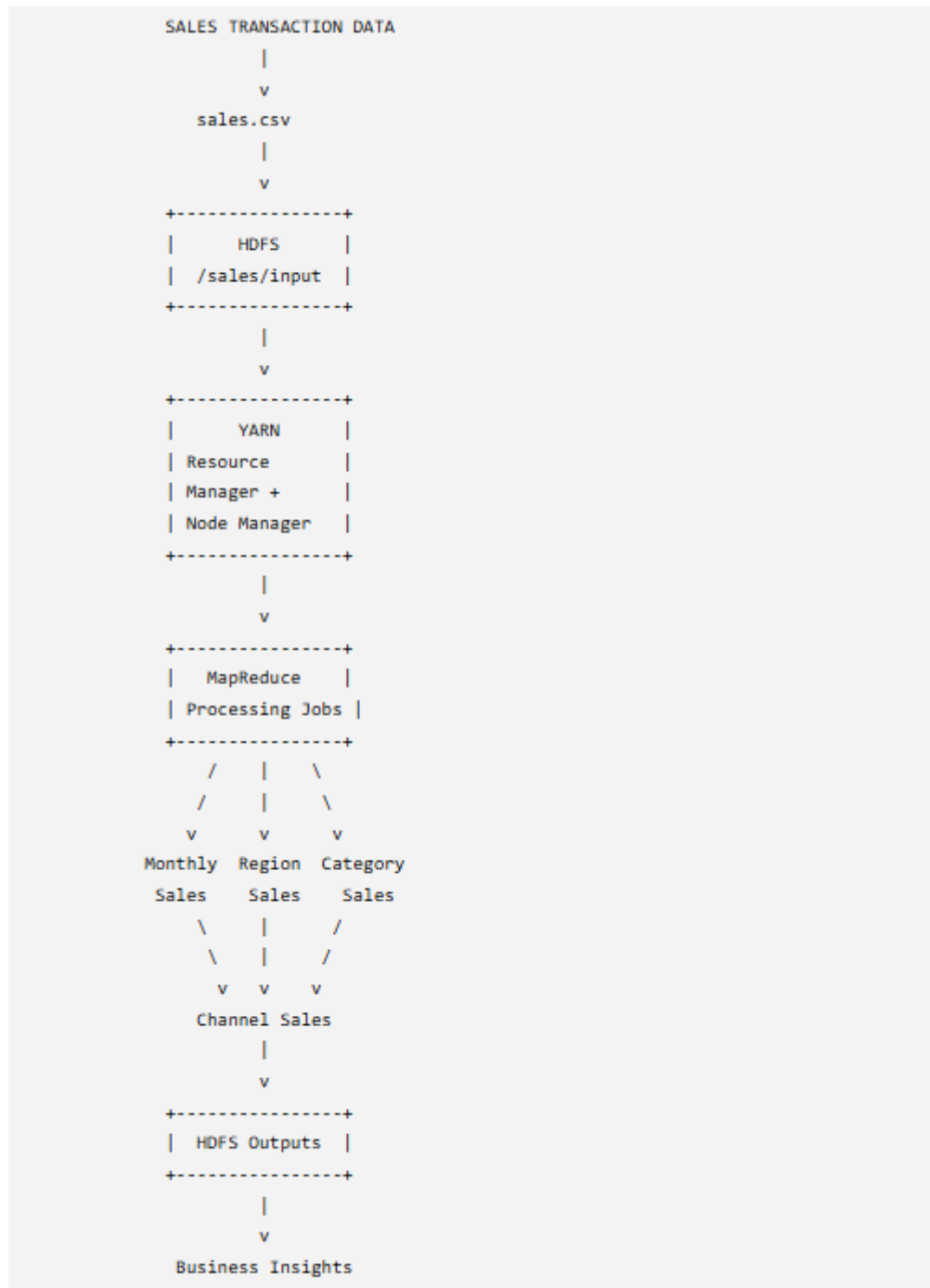
The dataset was uploaded to HDFS at:

/sales/input/sales.csv

The input file is approximately **779 KB** in the proof-of-concept environment.

2.2 Hadoop Architecture

The proposed architecture consists of the following components:



2.3 Processing Workflow

The implementation follows these steps:

Step 1 – Data Ingestion

The sales CSV file is prepared in the Linux environment and uploaded into HDFS.

/sales/input/sales.csv

Step 2 – Distributed Storage

HDFS is used to store the input data. This provides scalable and fault-tolerant storage for large datasets.

Step 3 – Distributed Processing

MapReduce programs are submitted through Hadoop and executed using YARN.

The project performs four major analyses:

- Monthly sales
- Regional sales
- Category sales
- Channel sales

Step 4 – Aggregation

The Mapper processes individual transaction records and generates key-value pairs based on the required analytical dimension.

The Reducer then aggregates the values to produce total sales for each month, region, category, or channel.

Step 5 – Output Storage

The results are stored in HDFS under separate output directories.

For example:

/sales/monthly-output

/sales/region-output

/sales/category-output

/sales/channel-output

Step 6 – Analysis

The resulting files are examined using HDFS commands such as:

```
hdfs dfs -cat
```

The results can then be interpreted to identify business trends and performance differences.

3. PROOF OF CONCEPT AND PRELIMINARY RESULTS

The proof of concept was successfully implemented using:

- **HDFS** – distributed storage
- **YARN** – resource and job management
- **MapReduce** – distributed data processing

The project processed a dataset containing **10,000 sales transactions**.

The MapReduce jobs successfully generated four analytical outputs.

3.1 Monthly Sales Analysis

The monthly MapReduce analysis produced the following results:

Month	Total Sales
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2026-01	80.57 M
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2026-02	76.12 M
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2026-03	83.52 M
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2026-04	84.41 M
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2026-05	87.74 M
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2026-06	81.34 M
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2026-07	88.08 M
---------	---------

2026-08	88.12 M
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2026-09	73.94 M
---------	---------

2026-10	91.01 M
---------	---------

2026-11	80.66 M
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2026-12	86.03 M
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The highest monthly sales occurred in **October 2026**, with approximately **91.01 million** in sales.

The lowest monthly sales occurred in **September 2026**, with approximately **73.94 million**.

This indicates noticeable month-to-month variation in sales performance and demonstrates how the Hadoop solution can be used for sales trend analysis.

3.2 Regional Sales Analysis

The regional analysis produced:

Region	Total Sales
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Central	164.35 M
---------	----------

East	167.09 M
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North	170.01 M
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Northeast	167.77 M
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Region	Total Sales
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South	165.86 M
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West	166.46 M
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The **North region** generated the highest sales, with approximately **170.01 million**.

The relatively close values among the regions also provide an opportunity for further analysis to understand the factors influencing regional performance.

3.3 Category Sales Analysis

The category analysis produced:

Category	Total Sales
----------	-------------

Accessories	45.20 M
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Electronics	753.87 M
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Furniture	132.47 M
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Office	70.00 M
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Electronics is the dominant category in the sample dataset, generating approximately **753.87 million** in sales.

This result could be used by business teams to investigate product demand, inventory planning, and category-level performance.

3.4 Channel Sales Analysis

The channel analysis produced:

Channel	Total Sales
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Distributor	325.62 M
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Online	343.68 M
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Store	332.24 M
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The **Online channel** generated the highest sales among the three channels, with approximately **343.68 million**.

This indicates that the online channel is an important contributor to sales in the sample dataset.

4. IMPLEMENTATION TIMELINE

Phase	Activity	Status
1	Identify business problem	Completed
2	Prepare sales dataset	Completed
3	Configure Hadoop environment	Completed
4	Upload data to HDFS	Completed
5	Implement monthly sales analysis	Completed
6	Implement regional sales analysis	Completed
7	Implement category analysis	Completed
8	Implement channel analysis	Completed
9	Validate outputs and capture screenshots	Completed
10	Prepare proposal and presentation	Current

5. EXPECTED BUSINESS IMPACT AND FUTURE SCOPE

The proposed Hadoop-based sales analytics solution can reduce the manual effort required to process and aggregate large sales datasets.

The monthly sales analysis can support:

- Sales trend monitoring
- Sales planning
- Forecasting
- Identification of high- and low-performing periods

Regional analysis can help identify geographical differences in business performance.

Category analysis can help organizations understand which product categories contribute most to revenue.

Channel analysis can help management compare Online, Store, and Distributor performance.

Future Scope

The current implementation is a proof of concept. A production implementation could be extended with:

1. Automated data ingestion.
2. Larger datasets containing millions of transactions.
3. Scheduled daily or hourly processing.
4. Data-quality validation.

5. Additional KPIs such as average order value and sales growth.
 6. Business dashboards for visualization.
 7. Integration with organizational databases.
 8. Near-real-time sales analytics.
 9. Predictive analytics for sales forecasting.
 10. Additional Hadoop ecosystem technologies where appropriate.
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6. CONCLUSION

This project demonstrates the practical application of Big Data technologies to a sales-analysis problem from a data analyst's perspective.

The proof of concept successfully uses **HDFS, YARN, and MapReduce** to store and process 10,000 sales transactions. The system generates monthly, regional, category-wise, and channel-wise sales results.

The analysis identified October as the highest-sales month, North as the highest-performing region, Electronics as the dominant category, and Online as the highest-performing sales channel in the sample dataset.

The project demonstrates that Hadoop can provide a scalable foundation for processing large sales datasets and generating business insights. The same architecture can be extended to significantly larger datasets and integrated with automated ingestion, dashboards, forecasting, and other advanced analytics capabilities.